



Harshita Sharma

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B.Tech - Chemical Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	7.5 (Current)	2024-Present
Senior Secondary	MPBSE Board	95.8%	2024
Secondary	CBSE Board	97.5%	2022

EXPERIENCE

- WorldQuant Brain** *Research Consultant* October. 2025 - present Remote
 - Utilized a **quantitative approach** to develop, backtest, and optimize **alpha trading strategies** using historical market data.
 - Engineered **60+ alpha strategies** with **Sharpe ratios exceeding 1.25** and **low pairwise correlation (< 0.7)** across multiple asset classes and geographies including the **USA, Europe, and Asia**, focusing on performance consistency and diversification
- AYUSH Upchar** *Data Analyst and Machine Learning Intern* August 2025 - present Remote
 - Performed **data cleaning** and preprocessing on healthcare product datasets using **Python (Pandas, NumPy)**, applying controlled vocabularies, regex, and normalization to structure data (brand, form, benefit), and designed a **product substitution model** using **Sentence-BERT embeddings** with **cosine similarity** to recommend alternatives.

PROJECTS

- IPO Tracker** *Finance and Economics Club IIT Guwahati, IIT Guwahati* January 2026 Website
 - Built a full-stack **IPO analytics platform** integrating a **React.js** frontend with a scalable **FastAPI** backend, transforming fragmented financial, GMP, and subscription data into real-time, actionable investment insights with high system efficiency, featuring an interactive dashboard with probabilistic scoring, sector-wise comparisons, and data-driven decision intelligence.
 - Designed a sophisticated **machine learning pipeline** using **Random Forest, XGBoost, and CatBoost** to capture complex non-linear relationships, combined with Logistic Regression-based classification and an advanced **NLP-driven sentiment analysis engine** to generate IPO scores, real-time sentiment signals, and investment recommendations while reducing speculative bias.
- Systematic Equity Strategy** *Kriti, IIT Guwahati* March. 2026 Github
 - Independently developed a systematic equity investment strategy combining **momentum** and **low-volatility** factors on **NSE500**, consistently achieving **17.5% CAGR** with strong returns (**Sharpe 0.86, Max Drawdown-17.5%**).
 - Designed and successfully implemented a end-to-end quantitative pipeline including data architecture, **regime filtering** (200DMA), and transaction-cost-aware execution, delivering **310%** backtested returns over **2010-2020 period**.
- Techniche Website** *Techniche(Technical Fest), IIT Guwahati* July. 2025 Website
 - Developed a comprehensive full-stack website using **React.js, Node.js** and an optimised **MySQL** database to handle activity across 10+ portals, significantly amplifying nationwide participation across **300+ cities** with seamless user experience.
 - Developed the backend for the **Aquawars** registration system using **Node.js, Express, and Sequelize**, ensuring efficient data handling, robust validations, secure database integration, and scalable API design high system efficiency.

TECHNICAL SKILLS

- Programming:** Python, C/C++, JavaScript, Node.js
- Libraries:** Numpy, Pandas, Matplotlib, Plotly, Seaborn
- Web Technologies:** HTML, CSS, JavaScript, React*
- Finance:** Algorithmic Trading*, Portfolio Optimization*

*Elementary proficiency

KEY COURSES TAKEN

- Mathematics:** Linear Algebra, Complex Analysis, Real Analysis
- DS & ML:** Summer Analytics, IITG.ai, Time Series, Deep Learning, SQL

ACHIEVEMENTS

- AlgoUniversity**, Selected for Tech Fellowship, ranking 126 among 100,000+ applicants. 2026
- WorldQuant Brain**, Recieved **gold medal** in WorldQuant Challenge 2025
- Codeforces**, Maximum rating **1416**, current rating **1322**
- MN Analyze, FEC IITG, PAN India rank 9** 2025

POSITIONS OF RESPONSIBILITY AND EXTRACURRICULARS

- Capital and Investment Head**, Finance and Economics Club, IIT Guwahati 2026 - Present
 - Leading initiatives in **corporate finance** and **financial modeling**, organizing structured learning sessions for students.
 - Driving student engagement in the **fintech domain** through strategic programs, workshops, and collaborative activities.
- Department Training representative**, CCD, coordinating training initiatives and activities. 2026 - Present
- Inter IIT Cultural meet**, Represented institute in the street play at Inter IIT Cultural Meet. 2024